MS1Vtx_barrel_l1ecal

overview · all regions and selections · rank 58 / 169

rank 58 / 169 by endcap / ht40 / D_equal

region	selection	D_equal	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg
endcap	all	0.324	0.320	0.331	0.328	0.337	0.321	0.307	0.326	34,970
endcap	ht40	0.405	0.401	0.411	0.410	0.418	0.402	0.388	0.407	7,390
endcap	bkg split	0.108	below 27,580 / above 7,390							
barrel	all	0.235	0.249	0.257	0.187	0.239	0.253	0.267	0.241	1,089
barrel	ht40	0.372	0.388	0.392	0.310	0.378	0.390	0.405	0.378	220
barrel	bkg split	0.176	below 869 / above 220							
all	all	0.293	0.295	0.307	0.280	0.308	0.299	0.292	0.295	36,059
all	ht40	0.377	0.379	0.389	0.364	0.390	0.382	0.376	0.379	7,610
all	bkg split	0.110	below 28,449 / above 7,610							

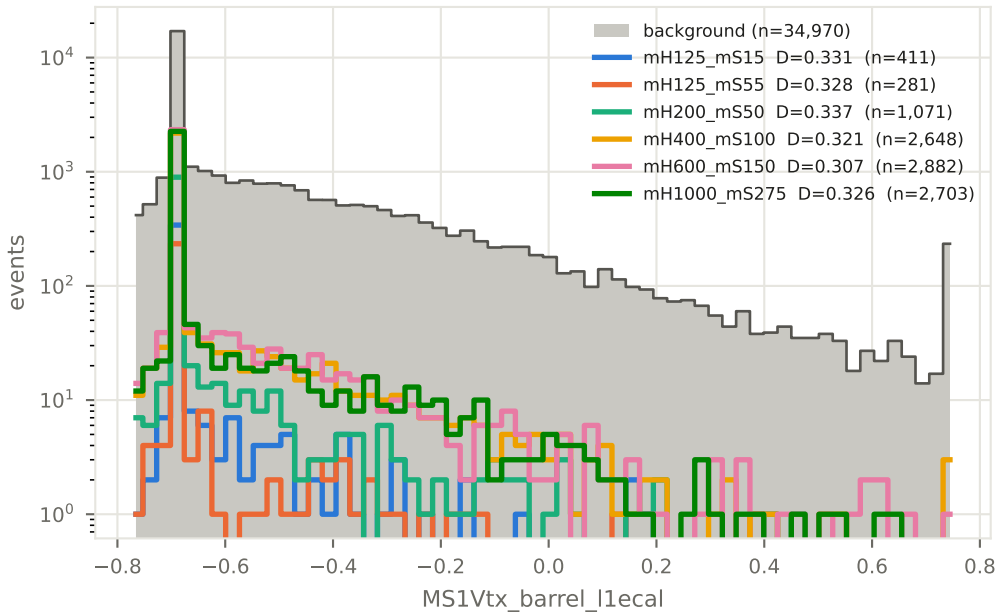
selections: all = every event; ht40 = htmissScalar_NOSYS >= 40 GeV, signal and background
pages: per region -> [all] 2x2, [ht40] 2x2, [background below vs above] 1x2

MS1Vtx_barrel_l1ecal

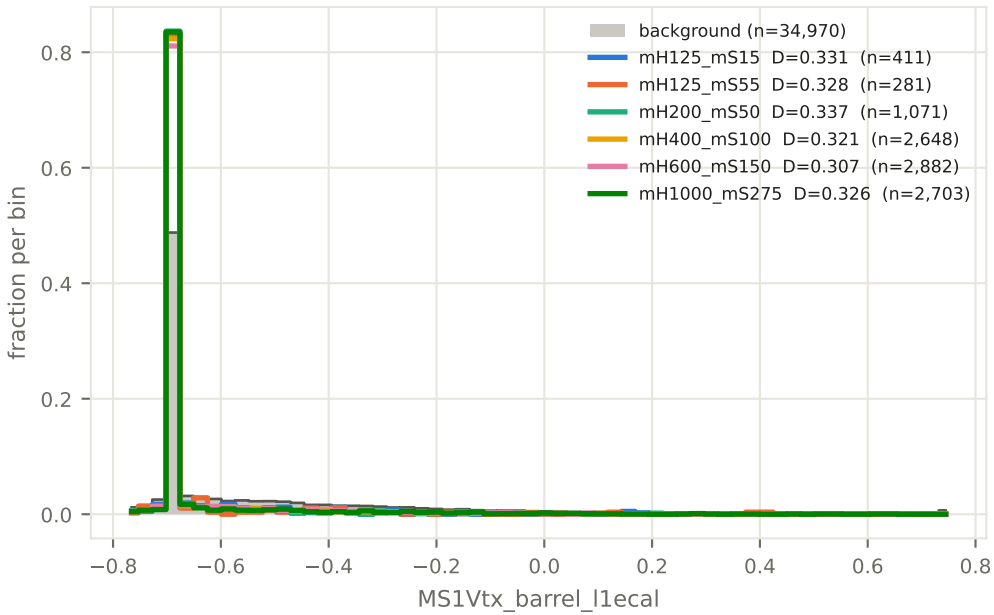
region: ENDCAP · selection: all (every event) · rank 58 / 169

D_equal 0.324 D_pooled 0.320 min 0.307 (mH600_mS150) max 0.337 (mH200_mS50) bkg n = 34,970 sig n = 9,996

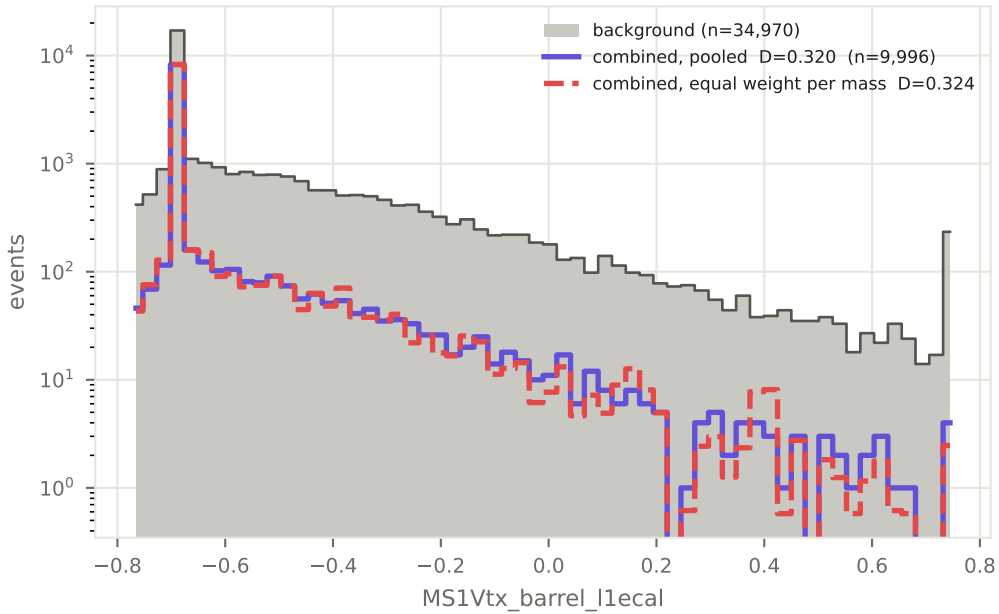
each mass point vs background · raw



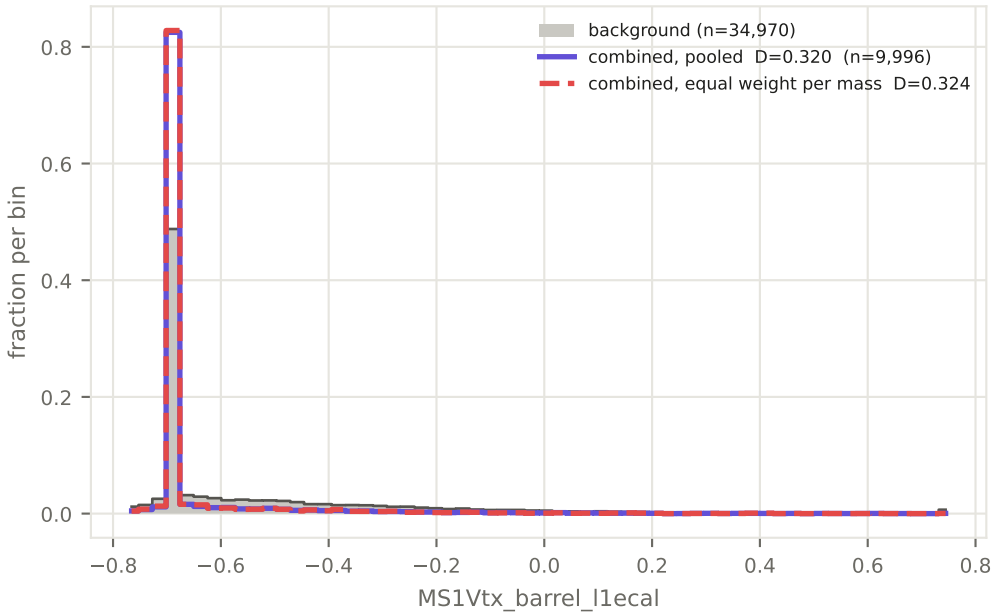
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



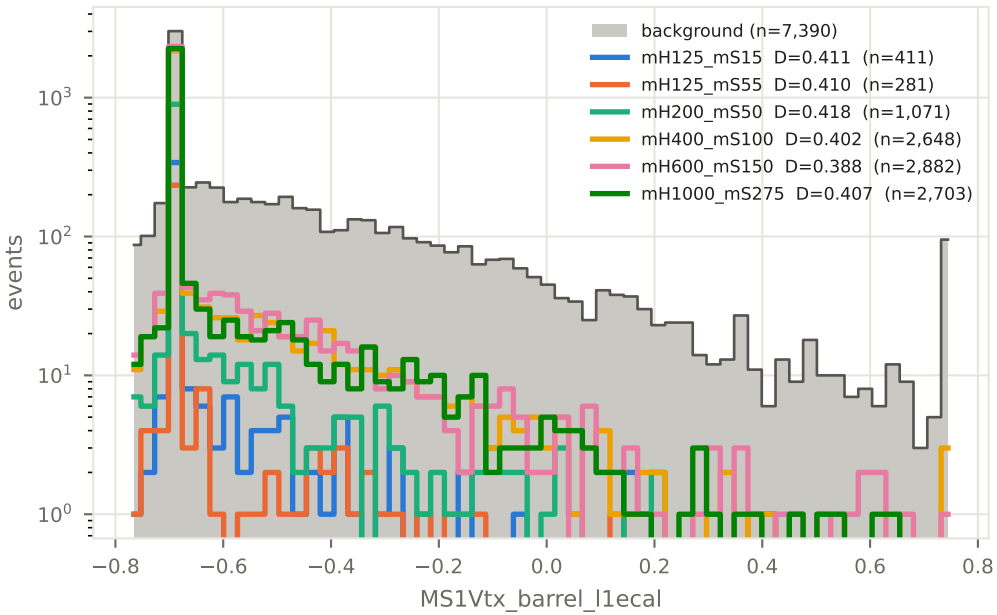
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_barrel_l1ecal

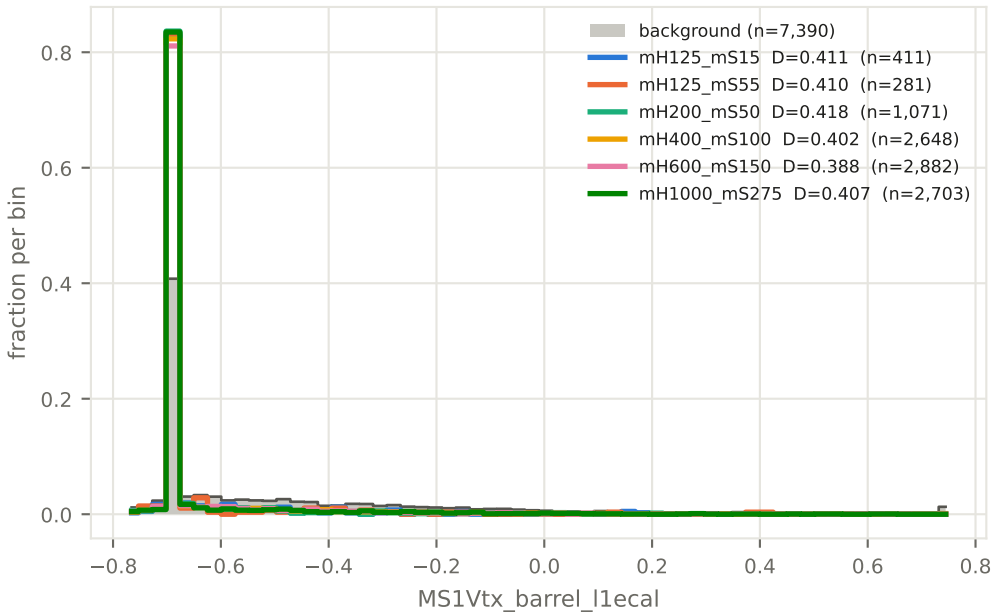
region: ENDCAP · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 58 / 169

D_equal 0.405 D_pooled 0.401 min 0.388 (mH600_mS150) max 0.418 (mH200_mS50) bkg n = 7,390 sig n = 9,996

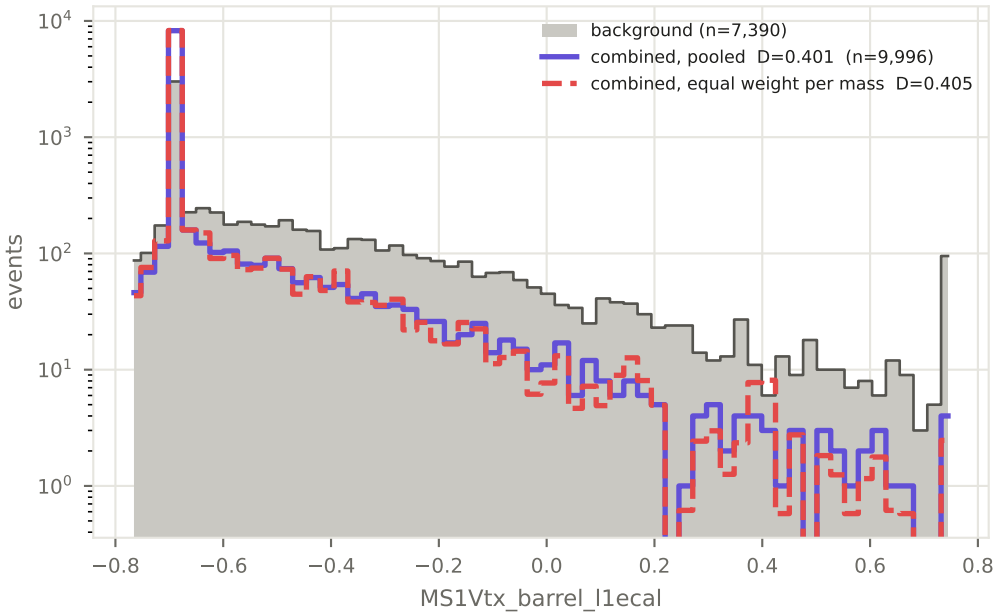
each mass point vs background · raw



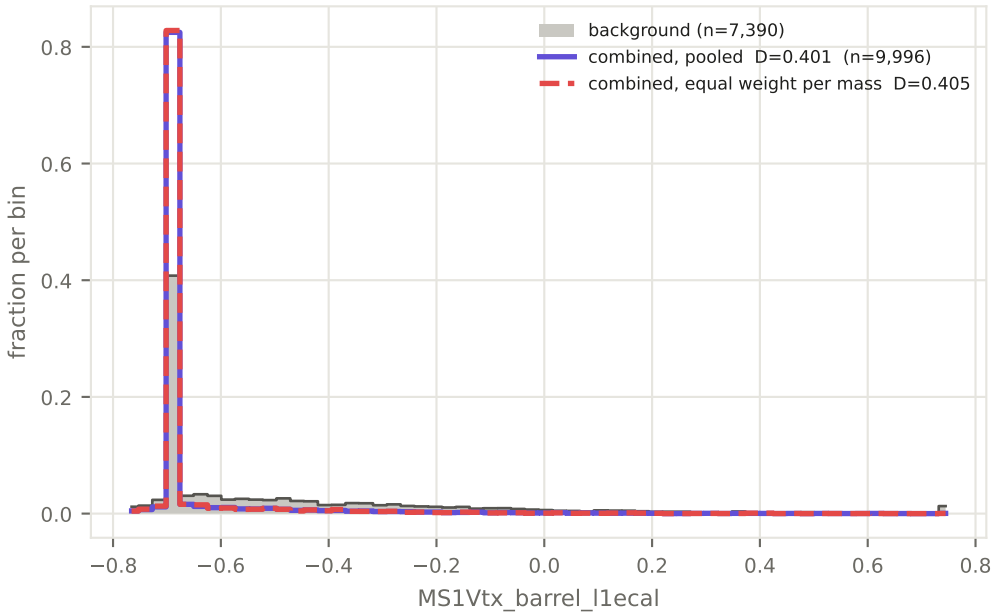
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

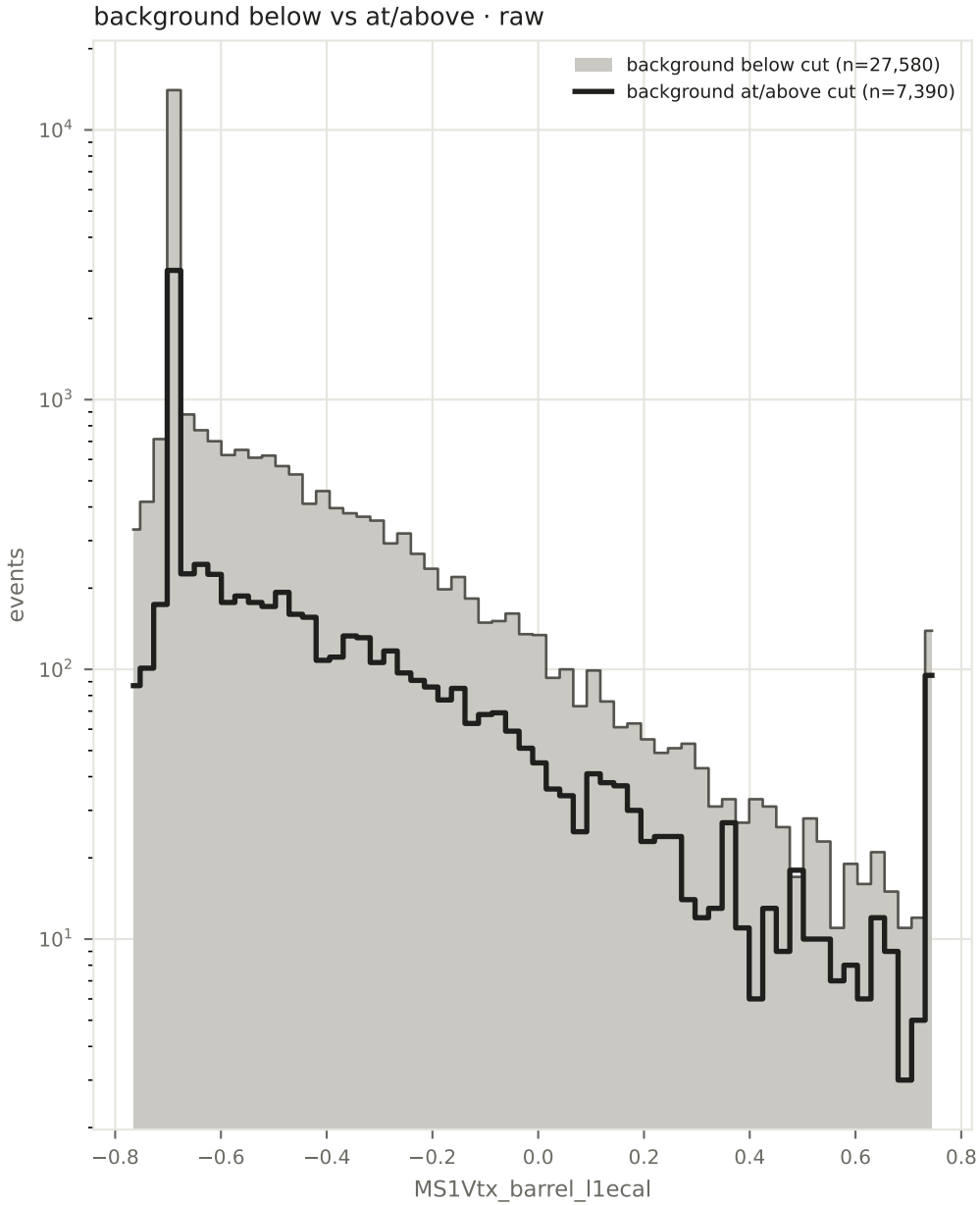


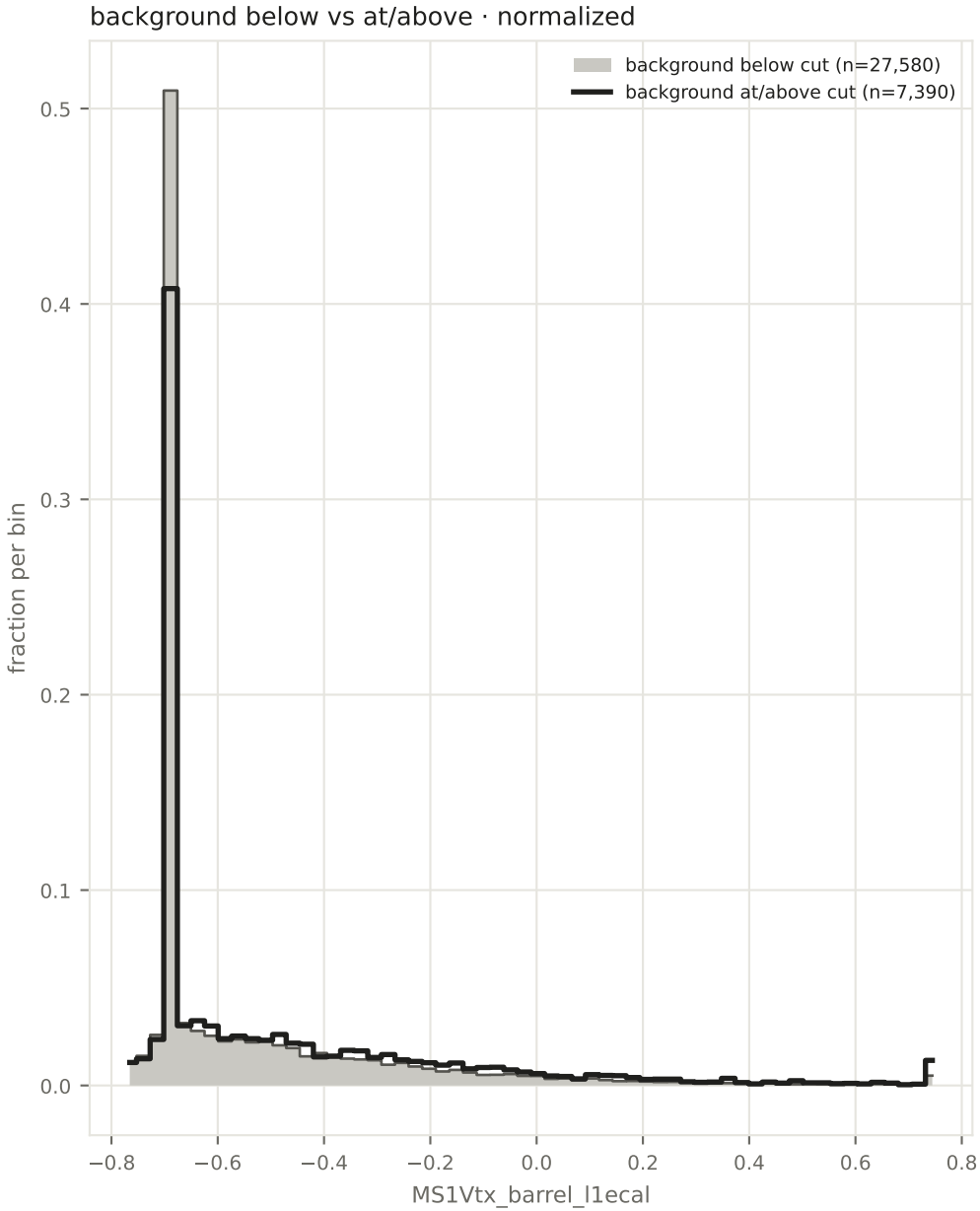
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_barrel_l1ecal

region: ENDCAP · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 58 / 169

D(below vs above) 0.108 n below = 27,580 n above = 7,390 large D = background shape changes across the cut



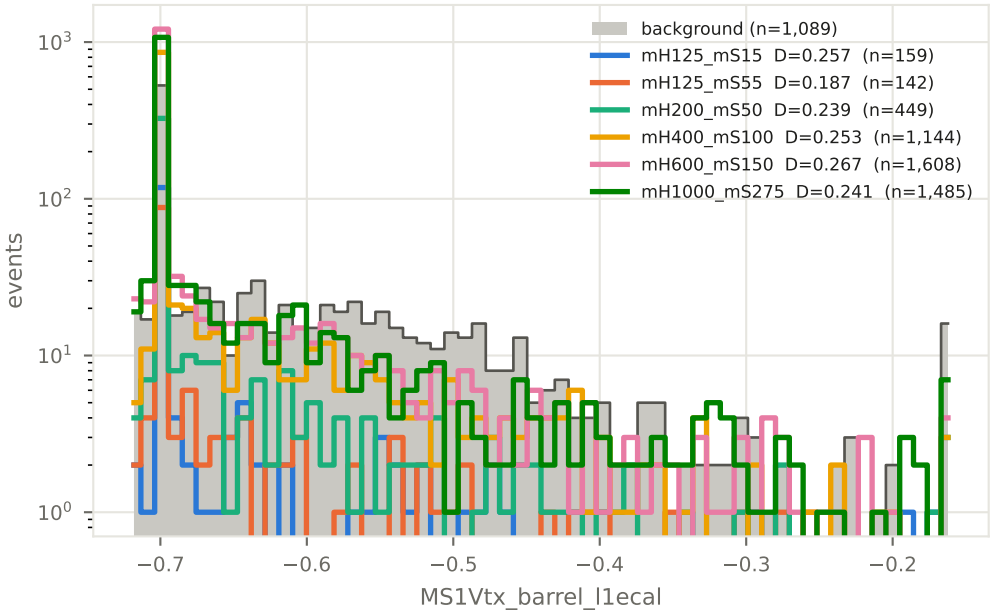


MS1Vtx_barrel_l1ecal

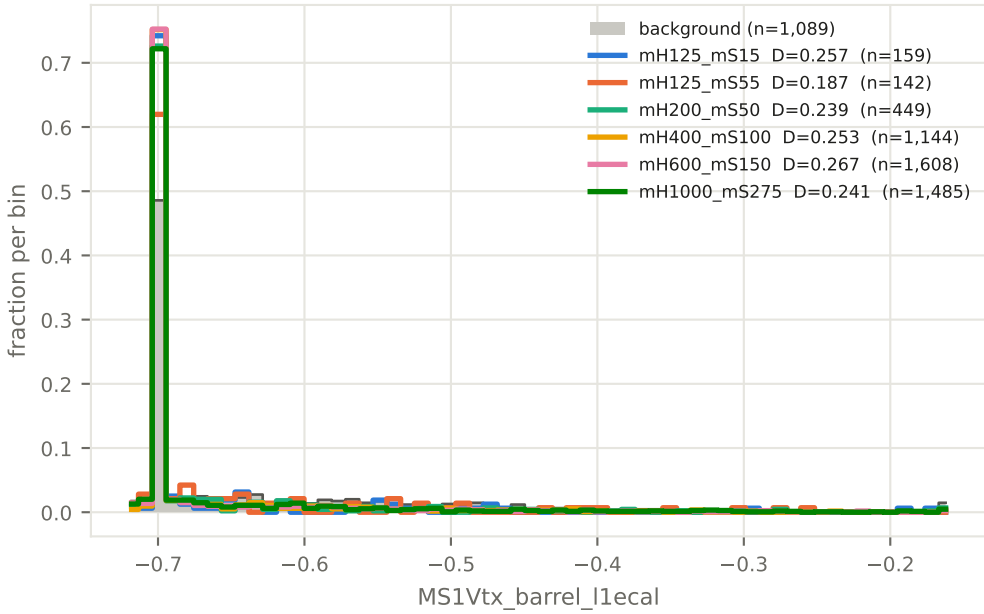
region: BARREL · selection: all (every event) · rank 58 / 169

D_equal 0.235 D_pooled 0.249 min 0.187 (mH125_mS55) max 0.267 (mH600_mS150) bkg n = 1,089 sig n = 4,987

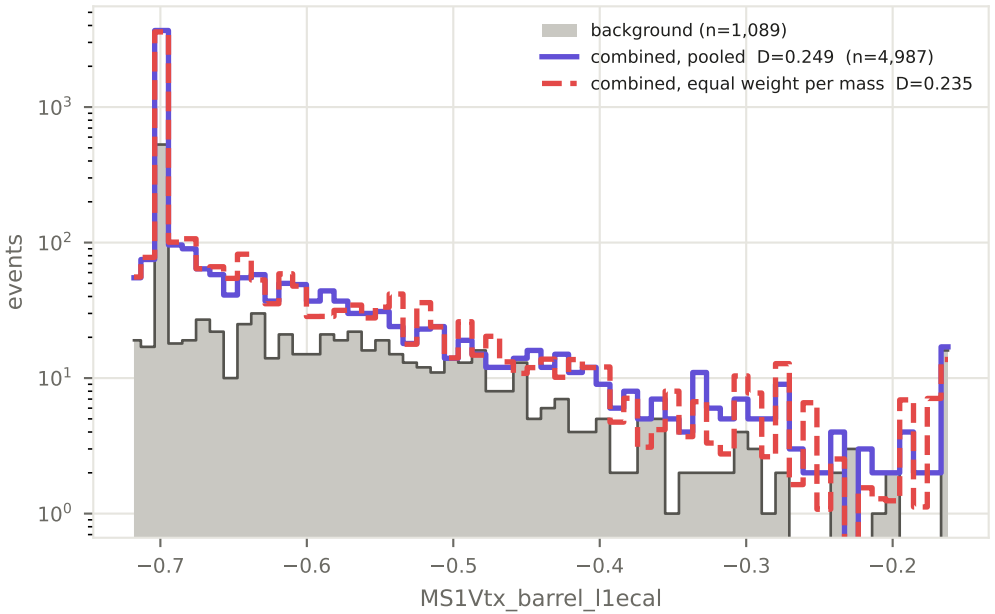
each mass point vs background · raw



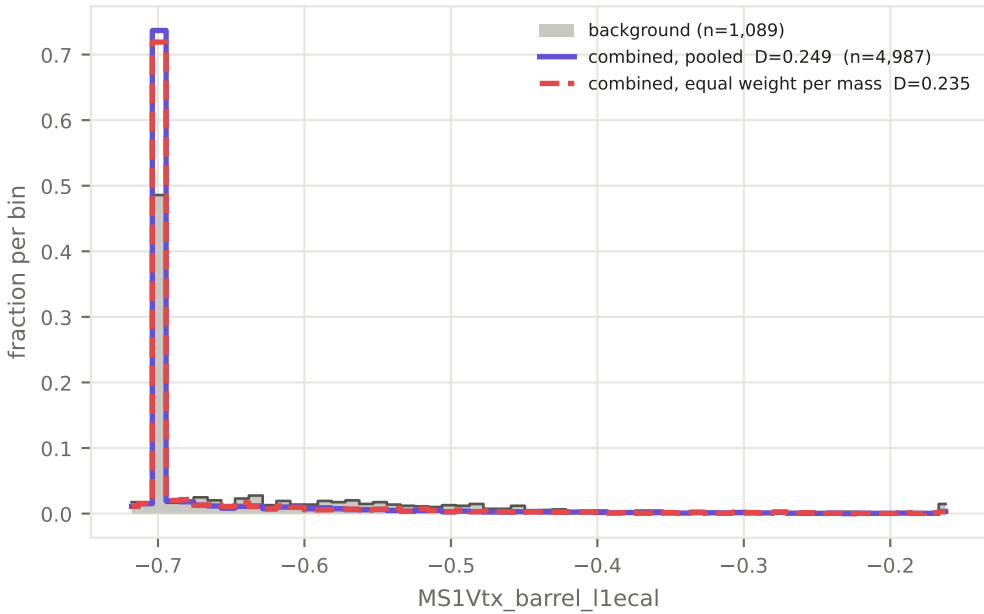
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



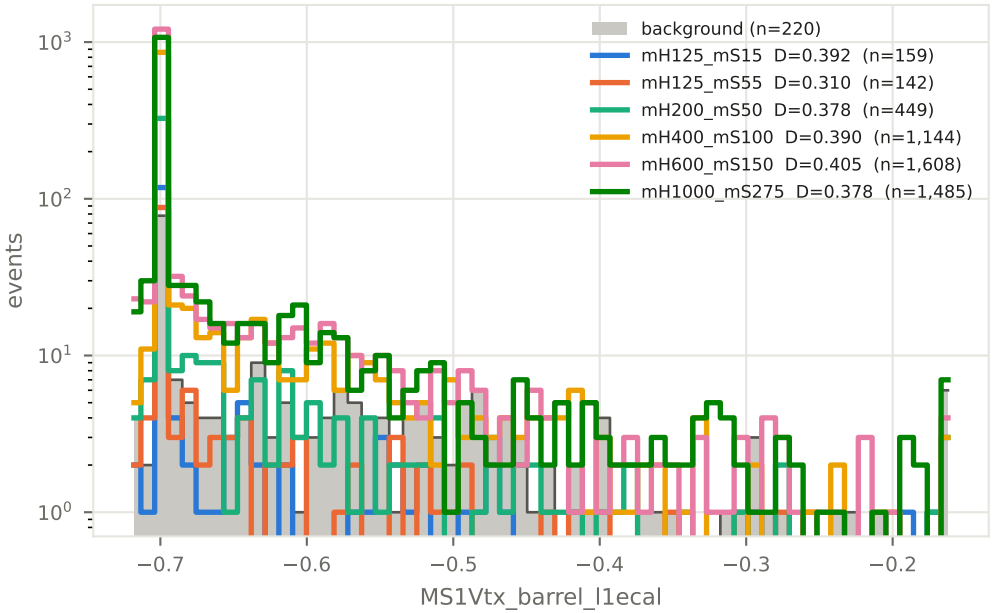
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_barrel_l1ecal

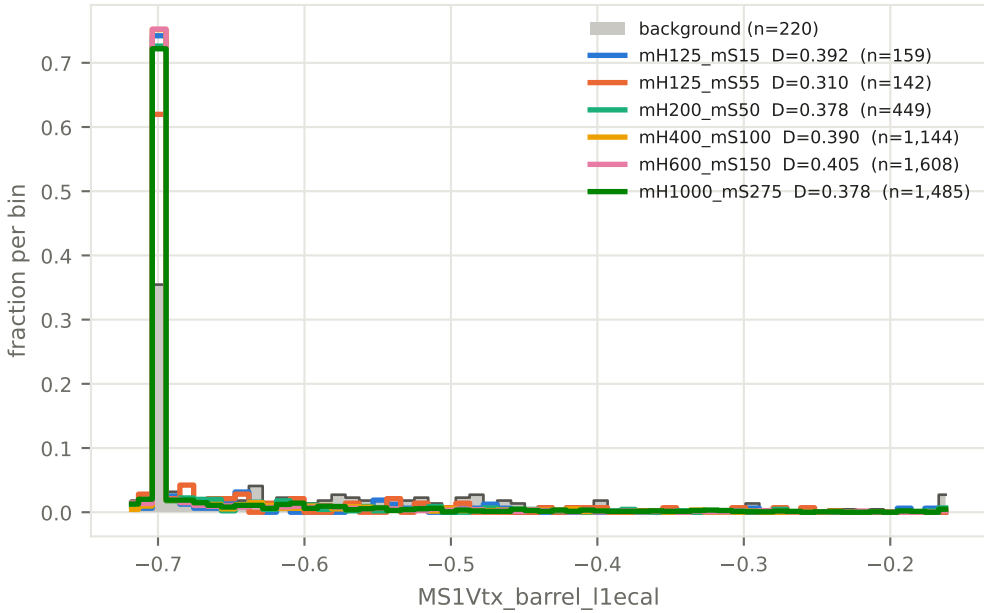
region: BARREL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 58 / 169

D_equal 0.372 D_pooled 0.388 min 0.310 (mH125_mS55) max 0.405 (mH600_mS150) bkg n = 220 sig n = 4,987

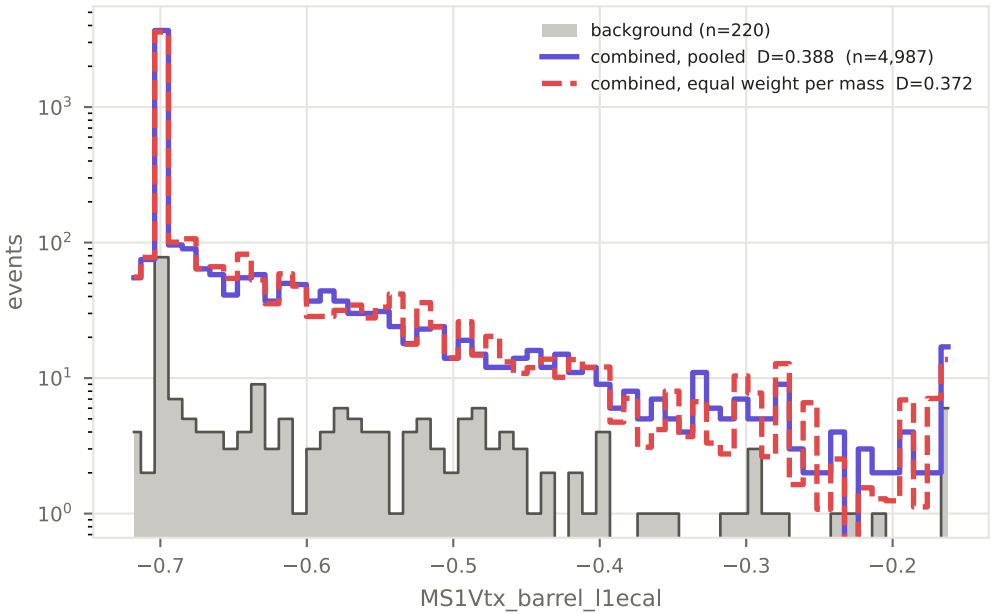
each mass point vs background · raw



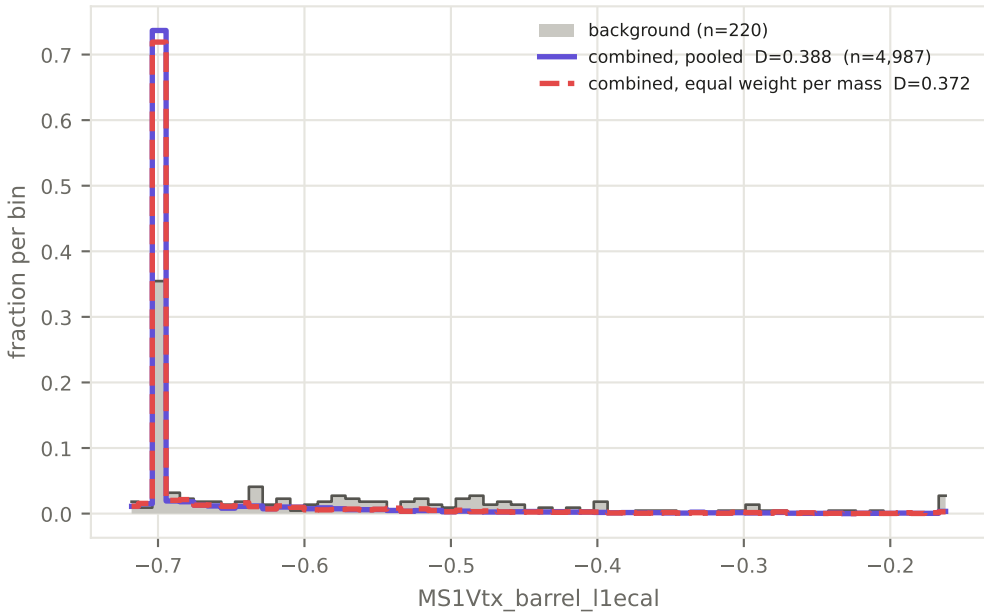
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

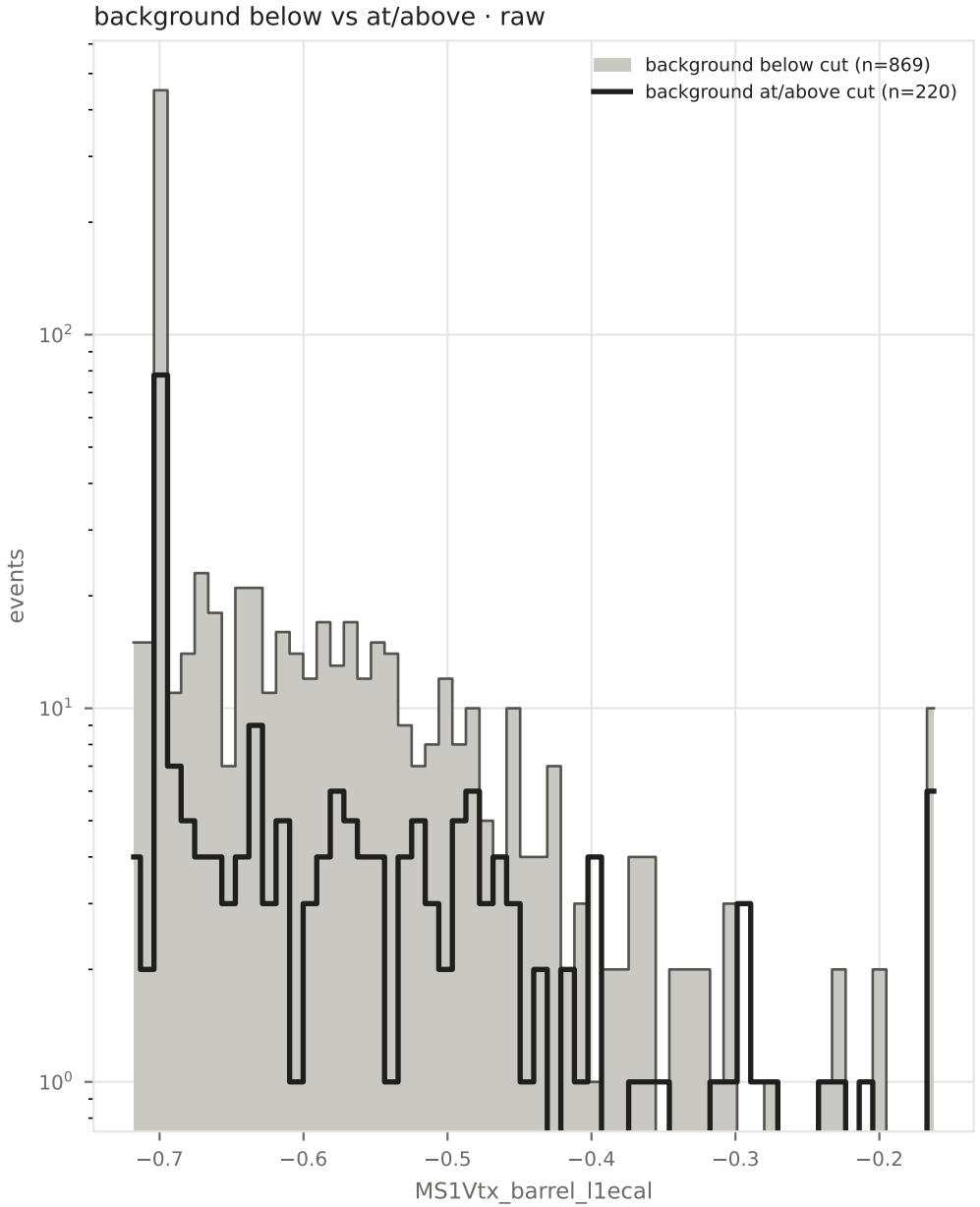


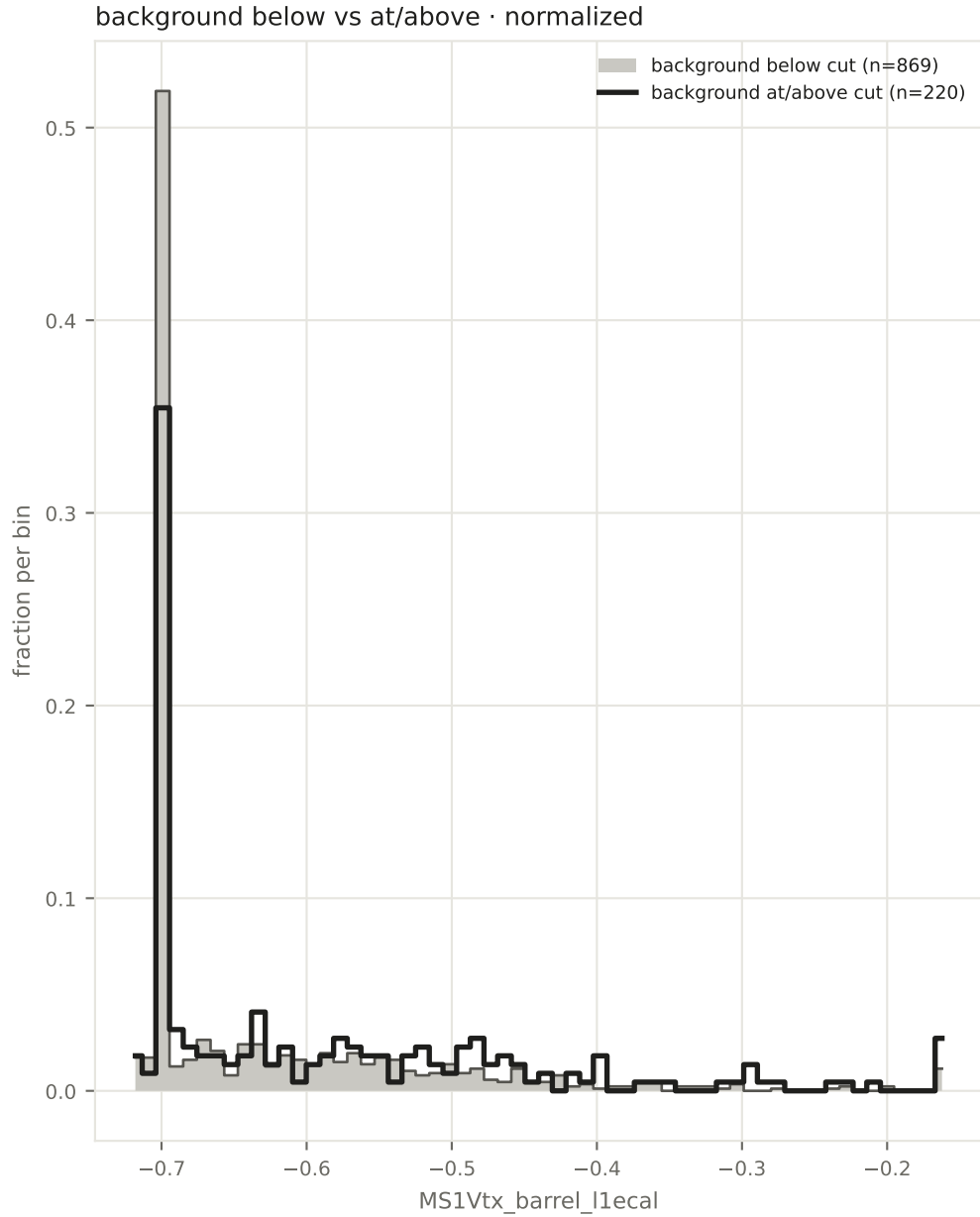
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_barrel_l1ecal

region: BARREL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 58 / 169

D(below vs above) 0.176 n below = 869 n above = 220 large D = background shape changes across the cut



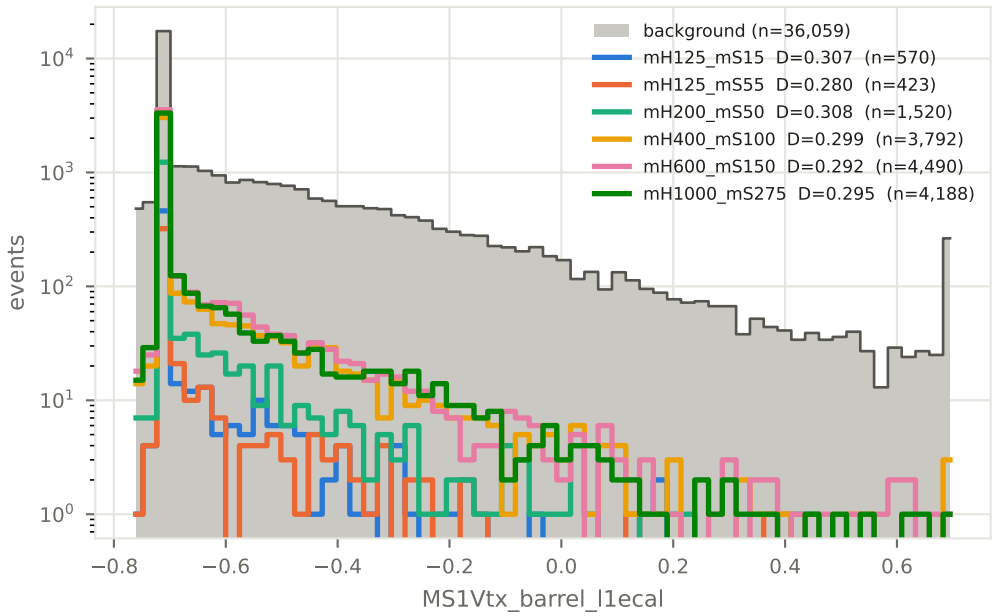


MS1Vtx_barrel_l1ecal

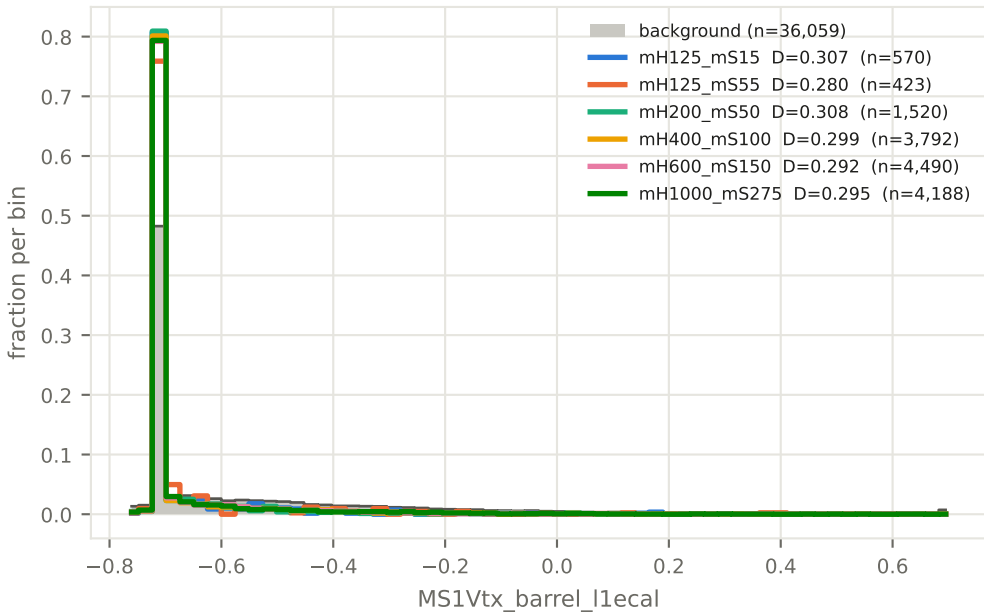
region: ALL · selection: all (every event) · rank 58 / 169

D_equal 0.293 D_pooled 0.295 min 0.280 (mH125_mS55) max 0.308 (mH200_mS50) bkg n = 36,059 sig n = 14,983

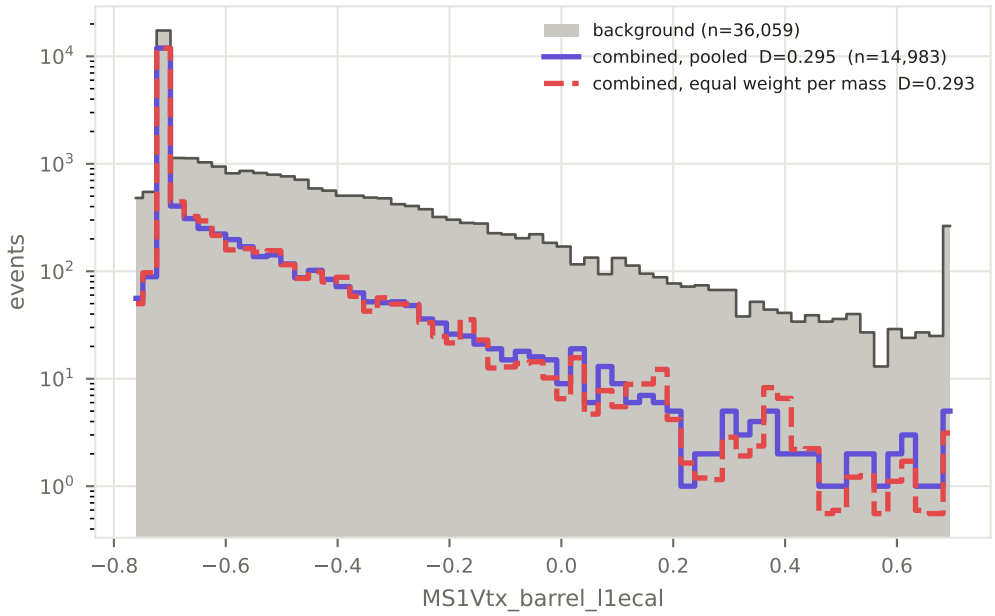
each mass point vs background · raw



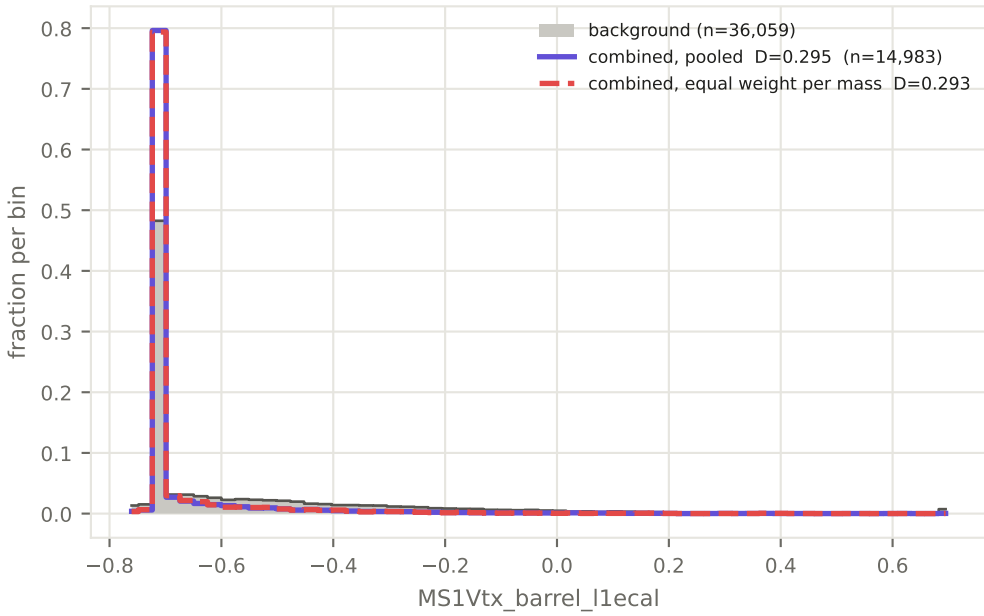
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



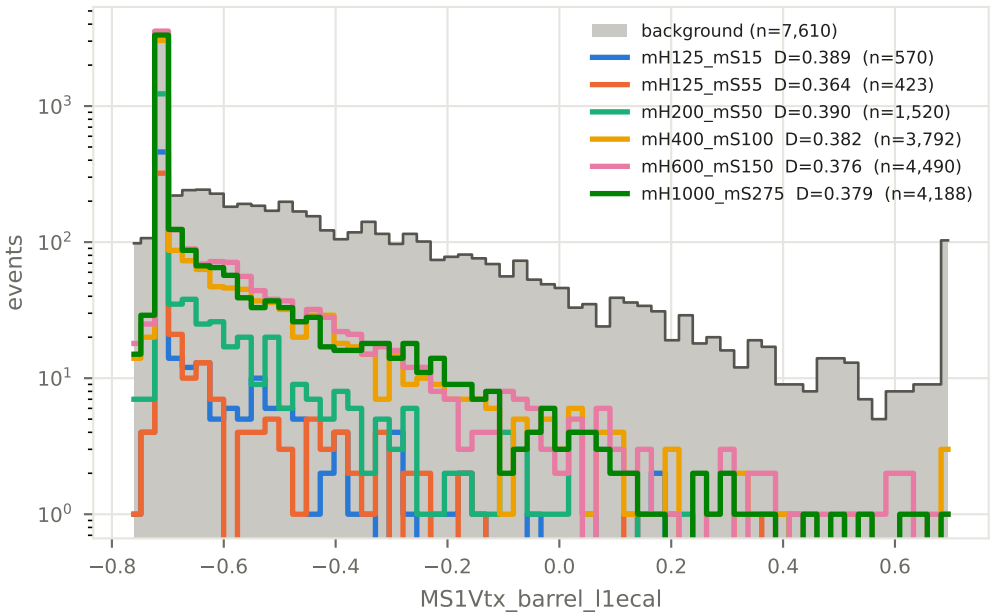
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_barrel_l1ecal

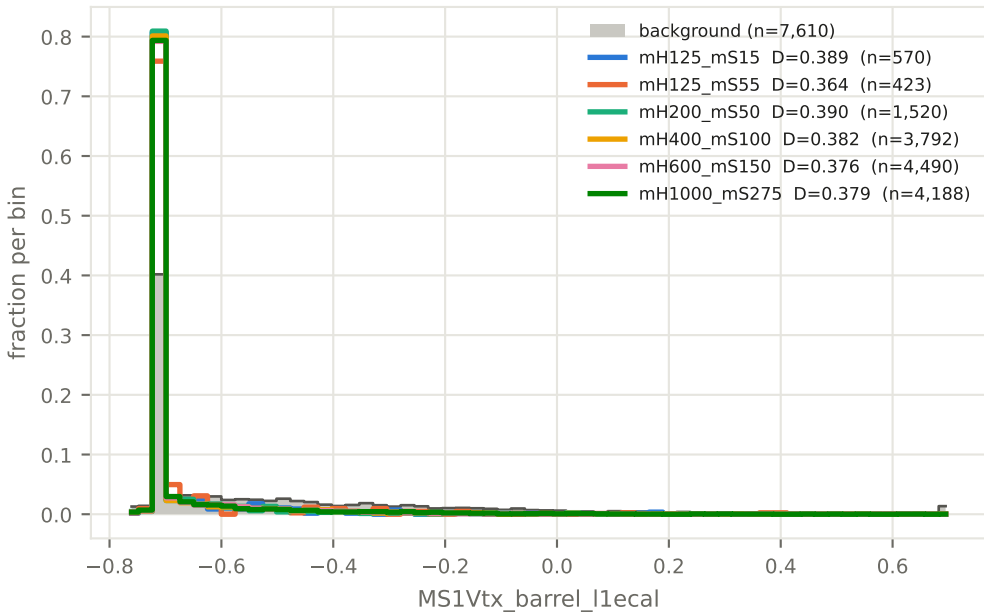
region: ALL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 58 / 169

D_equal 0.377 D_pooled 0.379 min 0.364 (mH125_mS55) max 0.390 (mH200_mS50) bkg n = 7,610 sig n = 14,983

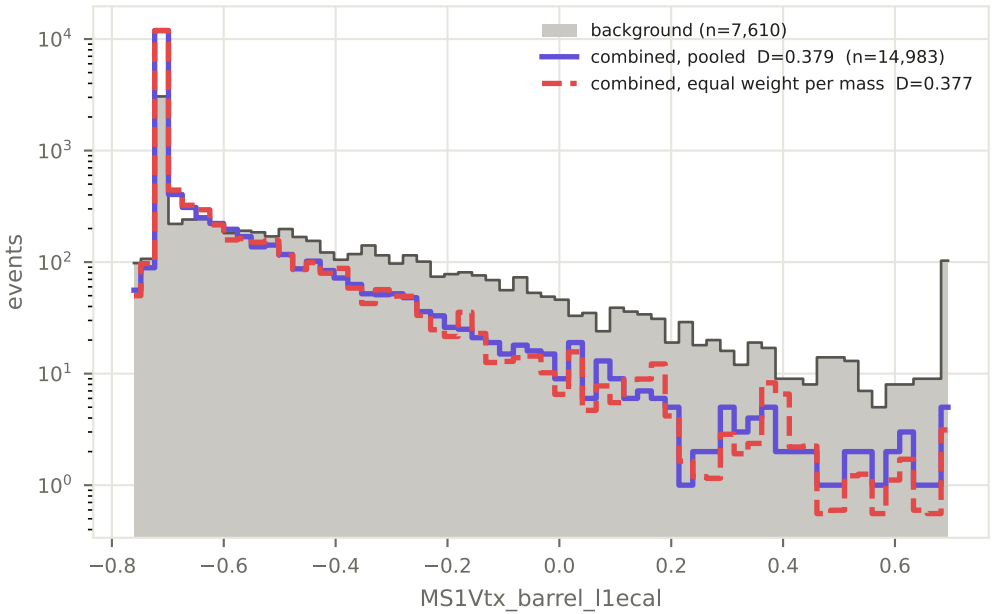
each mass point vs background · raw



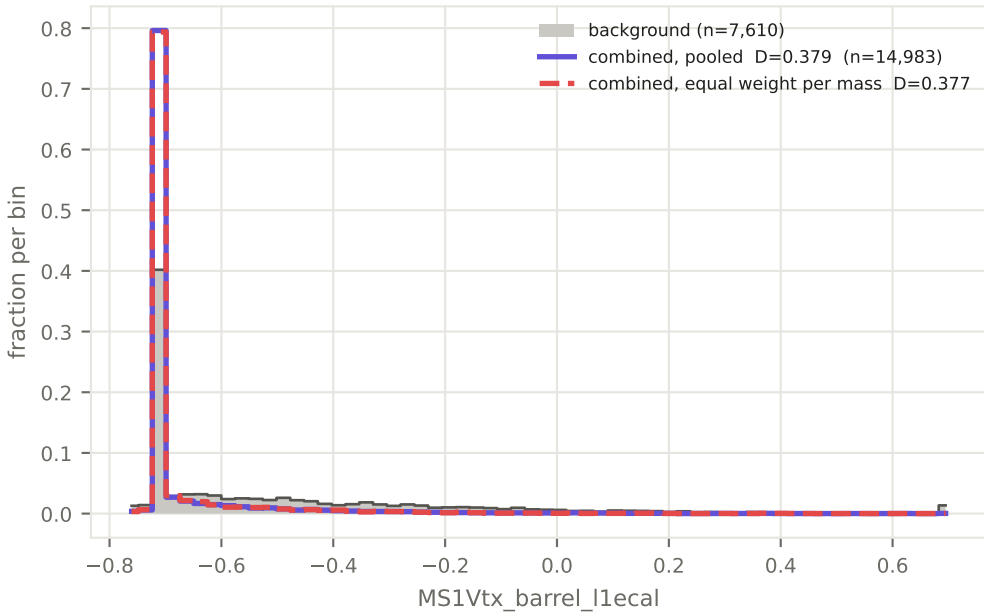
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



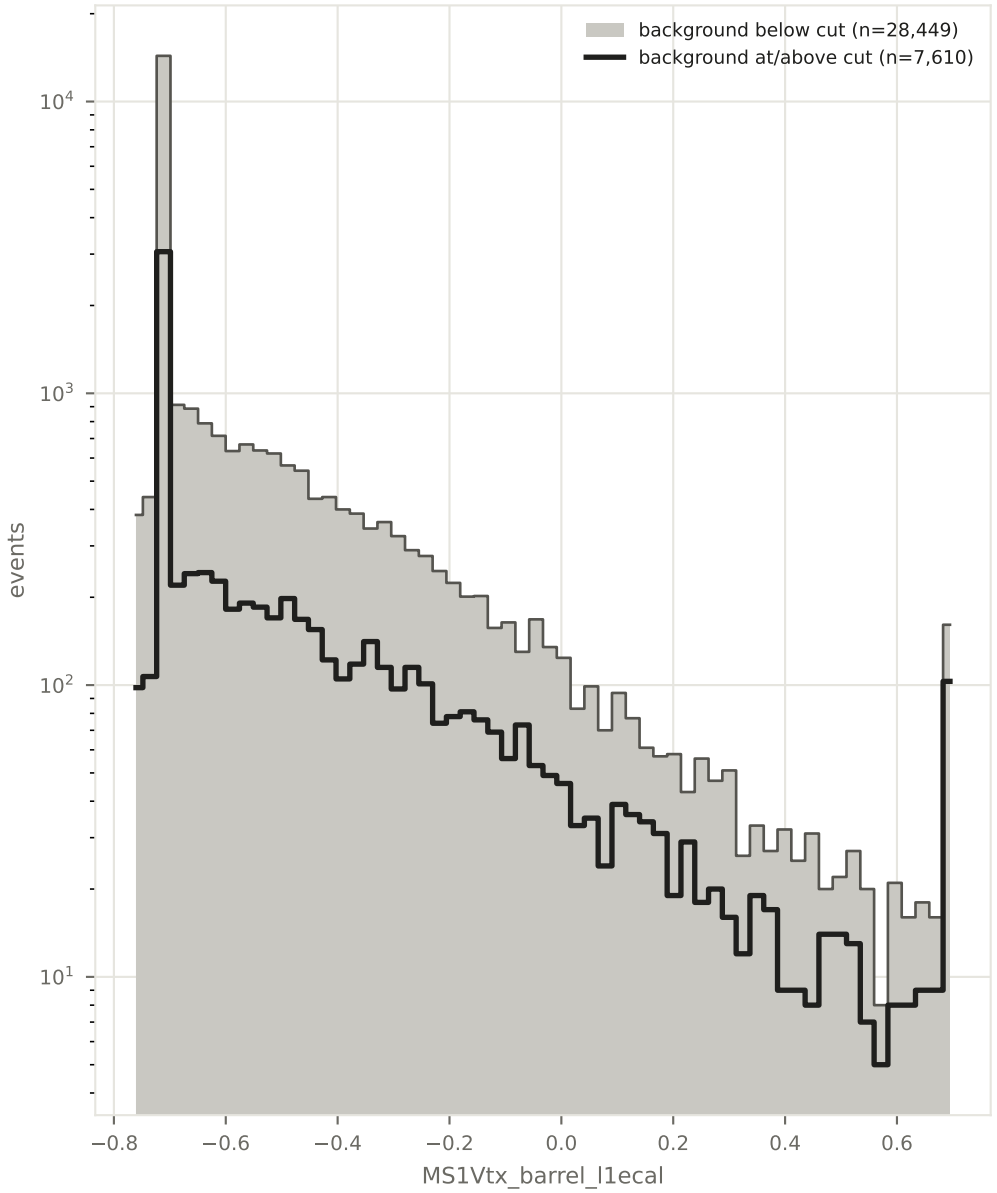
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_barrel_l1ecal

region: ALL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 58 / 169

D(below vs above) 0.110 n below = 28,449 n above = 7,610 large D = background shape changes across the cut

background below vs at/above · raw



background below vs at/above · normalized

